

SUSE Linux Enterprise for High-Performance Computing 15 SP4

Release Notes

SUSE Linux Enterprise for High-Performance Computing is a highly-scalable, high-performance open-source operating system designed to utilize the power of parallel computing. This document provides an overview of high-level general features, capabilities, and limitations of SUSE Linux Enterprise for High-Performance Computing 15 SP4 and important product updates.

These release notes are updated periodically. The latest version of these release notes is always available at <https://www.suse.com/releasenotes>. General documentation can be found at <https://documentation.suse.com/sle-hpc/15-SP4>.

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Contents

- 1 About the release notes 3
- 2 SUSE Linux Enterprise for High-Performance Computing 3
- 3 Modules, extensions, and related products 7
- 4 Technology previews 8
- 5 Modules 10
- 6 Changes affecting all architectures 11
- 7 Removed and deprecated features and packages 50
- 8 Obtaining source code 51

9 Legal notices **51**

A Changelog for 15 SP4 **53**

1 About the release notes

These Release Notes are identical across all architectures, and the most recent version is always available online at <https://www.suse.com/releasesnotes> .

Entries are only listed once but they can be referenced in several places if they are important and belong to more than one section.

Release notes usually only list changes that happened between two subsequent releases. Certain important entries from the release notes of previous product versions are repeated. To make these entries easier to identify, they contain a note to that effect.

However, repeated entries are provided as a courtesy only. Therefore, if you are skipping one or more service packs, check the release notes of the skipped service packs as well. If you are only reading the release notes of the current release, you could miss important changes.

2 SUSE Linux Enterprise for High-Performance Computing

SUSE Linux Enterprise for High-Performance Computing is a highly scalable, high performance open-source operating system designed to utilize the power of parallel computing for modeling, simulation and advanced analytics workloads.

SUSE Linux Enterprise for High-Performance Computing 15 SP4 provides tools and libraries related to High Performance Computing. This includes:

- Workload manager
- Remote and parallel shells
- Performance monitoring and measuring tools
- Serial console monitoring tool
- Cluster power management tool
- A tool for discovering the machine hardware topology
- System monitoring
- A tool for monitoring memory errors
- A tool for determining the CPU model and its capabilities (x86-64 only)

- User-extensible heap manager capable of distinguishing between different kinds of memory (x86-64 only)
- Serial and parallel computational libraries providing the common standards BLAS, LAPACK, ...
- Various MPI implementations
- Serial and parallel libraries for the HDF5 file format

2.1 Hardware Platform Support

SUSE Linux Enterprise for High-Performance Computing 15 SP4 is available for the Intel 64/AMD64 (x86-64) and AArch64 platforms.

2.2 Important Sections of This Document

If you are upgrading from a previous SUSE Linux Enterprise for High-Performance Computing release, you should review at least the following sections:

- *Section 2.4, “Support statement for SUSE Linux Enterprise for High-Performance Computing”*

2.3 Support and life cycle

SUSE Linux Enterprise for High-Performance Computing is backed by award-winning support from SUSE, an established technology leader with a proven history of delivering enterprise-quality support services.

SUSE Linux Enterprise for High-Performance Computing 15 has a 13-year life cycle, with 10 years of General Support and 3 years of Extended Support. The current version (SP4) will be fully maintained and supported until 6 months after the release of SUSE Linux Enterprise for High-Performance Computing 15 SP5.

Any release package is fully maintained and supported until the availability of the next release. Extended Service Pack Overlay Support (ESPOS) and Long Term Service Pack Support (LTSS) are also available for this product. If you need additional time to design, validate and test your upgrade plans, Long Term Service Pack Support (LTSS) can extend the support you get by an additional 12 to 36 months in 12-month increments, providing a total of 3 to 5 years of support on any given Service Pack.

For more information, see:

- The support policy at <https://www.suse.com/support/policy.html> 
- Long Term Service Pack Support page at <https://www.suse.com/support/programs/long-term-service-pack-support.html> 

2.4 Support statement for SUSE Linux Enterprise for High-Performance Computing

To receive support, you need an appropriate subscription with SUSE. For more information, see https://www.suse.com/support/programs/subscriptions/?id=SUSE_Linux_Enterprise_Server .

The following definitions apply:

L1

Problem determination, which means technical support designed to provide compatibility information, usage support, ongoing maintenance, information gathering and basic troubleshooting using available documentation.

L2

Problem isolation, which means technical support designed to analyze data, reproduce customer problems, isolate problem area and provide a resolution for problems not resolved by Level 1 or prepare for Level 3.

L3

Problem resolution, which means technical support designed to resolve problems by engaging engineering to resolve product defects which have been identified by Level 2 Support.

For contracted customers and partners, SUSE Linux Enterprise for High-Performance Computing is delivered with L3 support for all packages, except for the following:

- Technology Previews, see [Section 4, “Technology previews”](#)
- Sound, graphics, fonts and artwork
- Packages that require an additional customer contract, see [Section 2.4.1, “Software requiring specific contracts”](#)

SUSE will only support the usage of original packages. That is, packages that are unchanged and not recompiled.

2.4.1 Software requiring specific contracts

Certain software delivered as part of SUSE Linux Enterprise for High-Performance Computing may require an external contract. Check the support status of individual packages using the RPM metadata that can be viewed with `rpm`, `zypper`, or YaST.

2.4.2 Software under GNU AGPL

SUSE Linux Enterprise for High-Performance Computing 15 SP4 (and the SUSE Linux Enterprise modules) includes the following software that is shipped *only* under a GNU AGPL software license:

- Ghostscript (including subpackages)

SUSE Linux Enterprise for High-Performance Computing 15 SP4 (and the SUSE Linux Enterprise modules) includes the following software that is shipped under multiple licenses that include a GNU AGPL software license:

- MySpell dictionaries and LightProof
- ArgyllCMS

2.5 Documentation and other information

2.5.1 Available on the product media

- Read the READMEs on the media.
- Get the detailed change log information about a particular package from the RPM (where `FILENAME.rpm` is the name of the RPM):

```
rpm --changelog -qp FILENAME.rpm
```

- Check the `ChangeLog` file in the top level of the installation medium for a chronological log of all changes made to the updated packages.
- Find more information in the `docu` directory of the installation medium of SUSE Linux Enterprise for High-Performance Computing 15 SP4. This directory includes PDF versions of the SUSE Linux Enterprise for High-Performance Computing 15 SP4 Installation Quick Start Guide.

2.5.2 Online documentation

- For the most up-to-date version of the documentation for SUSE Linux Enterprise for High-Performance Computing 15 SP4, see <https://documentation.suse.com/sle-hpc/15-SP4>.
- Find a collection of White Papers in the SUSE Linux Enterprise for High-Performance Computing Resource Library at <https://www.suse.com/products/server#resources>.

3 Modules, extensions, and related products

This section comprises information about modules and extensions for SUSE Linux Enterprise for High-Performance Computing 15 SP4. Modules and extensions add functionality to the system.

3.1 Modules in the SLE 15 SP4 product line

The SLE 15 SP4 product line is made up of modules that contain software packages. Each module has a clearly defined scope. Modules differ in their life cycles and update timelines.

The modules available within the product line based on SUSE Linux Enterprise 15 SP4 at the release of SUSE Linux Enterprise for High-Performance Computing 15 SP4 are listed in the *Modules and Extensions Quick Start* at <https://documentation.suse.com/sles/15-SP3/html/SLES-all/article-modules.html>.

Not all SLE modules are available with a subscription for SUSE Linux Enterprise for High-Performance Computing 15 SP4 itself (see the column *Available for*).

For information about the availability of individual packages within modules, see <https://sc-c.suse.com/packages>.

3.2 Available extensions

The following extension is not covered by SUSE support agreements, available at no additional cost and without an extra registration key: SUSE Package Hub, see <https://package-hub.suse.com/>.

3.3 Related products

This section lists related products. Usually, these products have their own release notes documents that are available from <https://www.suse.com/releasesnotes>.

- SUSE Linux Enterprise Server: <https://www.suse.com/products/server>
- SUSE Linux Enterprise JeOS: <https://www.suse.com/products/server/jeos>
- SUSE Linux Enterprise Desktop: <https://www.suse.com/products/desktop>
- SUSE Linux Enterprise Server for SAP Applications: <https://www.suse.com/products/sles-for-sap>
- SUSE Linux Enterprise Real Time: <https://www.suse.com/products/realtime>
- SUSE Manager: <https://www.suse.com/products/suse-manager>

4 Technology previews

Technology previews are packages, stacks, or features delivered by SUSE which are not supported. They may be functionally incomplete, unstable or in other ways not suitable for production use. They are included for your convenience and give you a chance to test new technologies within an enterprise environment.

Whether a technology preview becomes a fully supported technology later depends on customer and market feedback. Technology previews can be dropped at any time and SUSE does not commit to providing a supported version of such technologies in the future.

Give your SUSE representative feedback about technology previews, including your experience and use case.

4.1 64K page size kernel flavor has been added

SUSE Linux Enterprise for High-Performance Computing for Arm 12 SP2 and later kernels have used a page size of 4K. This offers the widest compatibility also for small systems with little RAM, allowing to use Transparent Huge Pages (THP) where large pages make sense.

As a technology preview, SUSE Linux Enterprise for High-Performance Computing for Arm 15 SP4 adds a kernel flavor `64kb`, offering a page size of 64 KiB and physical/virtual address size of 52 bits. Same as the `default` kernel flavor, it does not use preemption.

Main purpose at this time is to allow for side-by-side benchmarking for High Performance Computing, Machine Learning and other Big Data use cases. Contact your SUSE representative if you notice performance gains for your specific workloads.



Important: Swap needs to be re-initialized

After booting the 64K kernel, any swap partitions need to be re-initialized to be usable. To do this, run the `swapon` command with the `--fixpgsz` parameter on the swap partition. Note that this process deletes data present in the swap partition (for example, suspend data). In this example, the swap partition is on `/dev/sdc1`:

```
swapon --fixpgsz /dev/sdc1
```



Important: Btrfs file system uses page size as block size

It is currently not possible to use Btrfs file systems across page sizes. Block sizes below page size are not yet supported and block sizes above page size might never be supported. During installation, change the default partitioning proposal and choose another file system, such as Ext4 or XFS, to allow rebooting from the default 4K page size kernel of the Installer into `kernel-64kb` and back.

See the *Storage Guide* for a discussion of supported file systems.



Warning: RAID 5 uses page size as stripe size

It is currently not yet possible to configure stripe size on volume creation. This will lead to sub-optimal performance if page size and block size differ.

Avoid RAID 5 volumes when benchmarking 64K vs. 4K page size kernels.

See the *Storage Guide* for more information on software RAID.



Note: Cross-architecture compatibility considerations

The SUSE Linux Enterprise for High-Performance Computing 15 SP4 kernels on x86-64 use 4K page size.

The SUSE Linux Enterprise for High-Performance Computing for POWER 15 SP4 kernel uses 64K page size.

5 Modules

5.1 HPC module

The HPC module contains HPC specific packages. These include the workload manager Slurm, the node deployment tool `clustduct`, `munge` for user authentication, the remote shell `mrsh`, the parallel shell `pdsh`, as well as numerous HPC libraries and frameworks.

This module is available with the SUSE Linux Enterprise for High-Performance Computing only. It is selected by default during the installation. It can be added or removed using the YaST UI or the `SUSEConnect` CLI tool. Refer to the system administration guide for further details.

5.2 NVIDIA Compute Module

The NVIDIA Compute Module provides the NVIDIA CUDA repository for SUSE Linux Enterprise 15. Note that any software within this repository is under a 3rd party EULA. For more information check <https://docs.nvidia.com/cuda/eula/index.html>.

This module is not selected for addition by default when installing SUSE Linux Enterprise for High-Performance Computing. It may be selected manually during installation from the *Extension and Modules* screen. You may also select it on an installed system using YaST. To do so, run from a shell as root `yast registration`, select: `Select Extensions` and search for `NVIDIA Compute Module` and press `Next`.

Important

Do not attempt to add this module with the SUSEConnect CLI tool. This tool is not yet capable of handling 3rd party repositories.

Once you have selected this module you will be asked to confirm the 3rd party license and verify the repository signing key.

6 Changes affecting all architectures

Information in this section applies to all architectures supported by SUSE Linux Enterprise for High-Performance Computing 15 SP4.

6.1 SLE HPC no longer a separate product

As of 15 SP4, SUSE Linux Enterprise for High-Performance Computing is no longer a separate product. As a result:

- the HPC Module can now be enabled in SUSE Linux Enterprise Server
- when migrating from SUSE Linux Enterprise for High-Performance Computing 15 SP3, SP4, and SP5, only SUSE Linux Enterprise Server 15 SP6 will be available as migration target. The result of such a migration will be an installation of SUSE Linux Enterprise Server with all the previously enabled modules.

6.2 Enriched system visibility in the SUSE Customer Center (SCC)

SUSE is committed to helping provide better insights into the consumption of SUSE subscriptions regardless of where they are running or how they are managed; physical or virtual, on-prem or in the cloud, connected to SCC or Repository Mirroring Tool (RMT), or managed by SUSE Manager. To help you identify or filter out systems in SCC that are no longer running or decommissioned, SUSEConnect now features a daily “ping”, which will update system information automatically. For more details see the documentation at <https://documentation.suse.com/subscription/suseconnect/single-html/SLE-suseconnect-visibility/>.

6.3 Automatically opened ports

Installing the following packages automatically opens the following ports:

- dolly - TCP ports 9997 and 9998
- slurm - TCP ports 6817, 6818, and 6819



Important

These release notes only document changes in SUSE Linux Enterprise for High-Performance Computing compared to the immediate previous service pack of SUSE Linux Enterprise for High-Performance Computing. The full changes and fixes can be found on the respective web site of the packages.

6.4 dolly

dolly has been updated to version 0.63.6. It includes some fixes for hostname resolution, a better documentation and now provides a default configuration for firewall.

6.5 memkind

memkind has been updated to version 1.12.0. The full list of changes is available at <http://memkind.github.io/memkind/> .

6.6 openblas

openblas has been updated to version 0.3.17. It contains performance regression fixes and optimization. For more information see <https://github.com/xianyi/OpenBLAS/releases/tag/v0.3.17> .

6.7 spack

6.7.1 v0.20.0

6.7.1.1 Features in this release

1. `requires()` directive and enhanced package requirements

We've added some more enhancements to requirements in Spack.

There is a new `requires()` directive for packages. `requires()` is the opposite of `conflicts()`. You can use it to impose constraints on this package when certain conditions are met. More details can be found [here \(https://spack.readthedocs.io/en/latest/build_settings.html#package-requirements\)](https://spack.readthedocs.io/en/latest/build_settings.html#package-requirements).

2. Exact versions

Spack did not previously have a way to distinguish a version if it was a prefix of some other version. For example, `@3.2` would match `3.2`, `3.2.1`, `3.2.2`, etc. You can now match *exactly* `3.2` with `@=3.2`. This is useful, for example, if you need to patch *only* the `3.2` version of a package. The new syntax is described in [the docs \(https://spack.readthedocs.io/en/latest/basic_usage.html#version-specifier\)](https://spack.readthedocs.io/en/latest/basic_usage.html#version-specifier).

Generally, when writing packages, you should prefer to use ranges like `@3.2` over the specific versions, as this allows the concretizer more leeway when selecting versions of dependencies. More details and recommendations are in the [packaging guide \(https://spack.readthedocs.io/en/latest/packaging_guide.html#ranges-versus-specific-versions\)](https://spack.readthedocs.io/en/latest/packaging_guide.html#ranges-versus-specific-versions).

3. More stable concretization

- Now, `spack concretize` will *only* concretize the new portions of the environment and will not change existing parts of an environment unless you specify `--force`. This has always been true for `unify:false`, but not for `unify:true` and `unify:when_possible` environments. Now it is true for all of them.
- The concretizer has a new `--reuse-deps` argument that *only* reuses dependencies. That is, it will always treat the *roots* of your environment as it would with `--fresh`. This allows you to upgrade just the roots of your environment while keeping everything else stable.

4. Specs in buildcaches can be referenced by hash.

- Previously, you could run `spack buildcache list` and see the hashes in build-caches, but referring to them by hash would fail.
- You can now run commands like `spack spec` and `spack install` and refer to buildcache hashes directly, e.g. `spack install /abc123`

5. New package and buildcache index websites

Our public websites for searching packages have been completely revamped and updated. You can check them out here:

- *Package Index*: <https://packages.spack.io> ↗
- *Buildcache Index*: <https://cache.spack.io> ↗

Both are searchable and more interactive than before. Currently major releases are shown; UI for browsing `develop` snapshots is coming soon.

6. Default CMake and Meson build types are now Release

Spack has historically defaulted to building with optimization and debugging, but packages like `llvm` can be enormous with debug turned on. Our default build type for all Spack packages is now `Release`. This has a number of benefits: * much smaller binaries; * higher default optimization level; and * defining `NDEBUG` disables assertions, which may lead to further speedups.

+ You can still get the old behavior back through requirements and package preferences.

6.7.1.2 Other new commands and directives

- `spack checksum` can automatically add new versions to package
- new command: `spack pkg grep` to easily search package files
- New `maintainers` directive
- Add `spack buildcache push` (alias to `buildcache create`)
- Allow using `-j` to control the parallelism of concretization
- Add `--exclude` option to ``spack external find'`

6.7.1.3 Other new features of note

- editing: add higher-precedence `SPACK_EDITOR` environment variable
- Many YAML formatting improvements from updating `ruamel.yaml` to the latest version supporting Python 3.6. .
- Requirements and preferences should not define (non-git) versions
- Environments now store spack version/commit in `spack.lock`
- User can specify the name of the `packages` subdirectory in repositories
- Add container images supporting RHEL alternatives
- make `version(...)` kwargs explicit

6.7.1.4 Notable refactors

- `buildcache create`: reproducible tarballs
- Bootstrap most of Spack dependencies using environments
- Split `satisfies(..., strict=True/False)` into two functions
- `spack install`: simplify behavior when inside environments

6.7.1.5 Removals, Deprecations, and disablements

- Module file generation is disabled by default; you'll need to enable it to use it
- Support for Python 2 was deprecated in `v0.19.0` and has been removed. `v0.20.0` only supports Python 3.6 and higher.
- Deprecated target names are no longer recognized by Spack. Use generic names instead:
 - `graviton` is now `cortex_a72`
 - `graviton2` is now `neoverse_n1`
 - `graviton3` is now `neoverse_v1`
- `blacklist` and `whitelist` in module configuration were deprecated in `v0.19.0` and are removed in this release. Use `exclude` and `include` instead.

- The `ignore=` parameter of the `extends()` directive has been removed. It was not used by any builtin packages and is no longer needed to avoid conflicts in environment views.
- Support for the old YAML buildcache format has been removed. It was deprecated in `v0.19.0`.
- `spack find --bootstrap` has been removed. It was deprecated in `v0.19.0`. Use `spack --bootstrap find` instead.
- `spack bootstrap trust` and `spack bootstrap untrust` are now removed, having been deprecated in `v0.19.0`. Use `spack bootstrap enable` and `spack bootstrap disable`.
- The `--mirror-name`, `--mirror-url`, and `--directory` options to `buildcache` and `mirror` commands were deprecated in `v0.19.0` and have now been removed. They have been replaced by positional arguments.
- Deprecate `env:` as top level environment key
- deprecate `buildcache create --rel`, `buildcache install --allow-root`
- Support for very old perl-like spec format strings (e.g., `$_$@$%@$+$+$=`) has been removed. This was deprecated in `v0.15`.

6.7.1.6 Notable Bugfixes

- bugfix: don't fetch package metadata for unknown concrete specs
- Improve package source code context display on error
- Relax environment manifest filename requirements and lockfile identification criteria
- `installer.py`: drop build edges of installed packages by default
- Bugfix: package requirements with git commits
- Package requirements: allow single specs in requirement lists
- conditional variant values: allow boolean
- `spack uninstall`: follow run/link edges on `--dependents`

For details, check the [upstream release notes \(https://github.com/spack/spack/releases/tag/v0.20.0\)](https://github.com/spack/spack/releases/tag/v0.20.0).

6.7.1.7 A script to set `LD_LIBRARY_PATH` is now provided

The command `spack load <target>` no longer sets the `LD_LIBRARY_PATH` environment variable. Since Spack was setting this variable to all libraries of the entire dependency stack, this has caused issues with system programs if they use a shared library that has been rebuilt by spack in a different way than the one provided by the system. In context of Spack this is not considered to be an issue as any Spack- built binary or library uses `RPATH` to set the location of shared libraries they depend on. Since Spack is used to build full solution stacks (including the final application binary this is not a problem.

If Spack is used to build a library stack for an application that is to be built outside of Spack, this is a problem. To remedy this, we provide the script `spack_get_libs.sh`. When called with a list of Spack packages, it will print shell commands to set and export `LD_LIBRARY` path prepended with with the path to the libraries from the Spack packages listed. The default shell is `bash`. With the option `--csh` a `csh` command line will be printed instead:

```
spack_get_libs.sh [--help] [--csh] lib ...
```

On `bash`, when the script is sourced, the environment updated directly. Additionally, the script will print settings for (or set) variables identifying include and library paths for each package of the form:

```
LIB_<PACKAGE_NAME>  
INC_<PACKAGE_NAME>
```

`<PACKAGE_NAME>` denotes the package name in upper case. These variables can be used during building.

6.7.2 v0.23.1

- Fix a correctness issue of `ArchSpec.intersects`.
- Make `extra_attributes` order independent in Spec hashing.
- Fix issue where system proxy settings were not respected in OCI build caches.
- Fix an issue where the `--test` concretizer flag was not forwarded correctly.
- Fix an issue where `codesign` and `install_name_tool` would not preserve hardlinks on Darwin.
- Fix an issue on Darwin where `codesign` would run on unmodified binaries.

- Patch configure scripts generated with `libtool < 2.5.4`, to avoid redundant flags when creating shared libraries on Darwin.
- Ensure proper UTF-8 encoding/decoding in logging.
- Fix issues related to `filter_file`.
- Fix an issue related to creating bootstrap source mirrors.
- Fix an issue where command line config arguments were not always top level.
- Fix an incorrect typehint of `concretized()`.
- Improve mention of next Spack version in warning. Tests: fix forward compatibility with Python 3.13.
- Docs: encourage use of `--oci-username-variable` and `--oci-password-variable`.
- Docs: ensure Getting Started has bootstrap list output in correct place.

6.7.3 v0.23.0

This is a new major release.

6.7.3.1 Features in this Release

- Spec splicing To make binary installation more seamless in Spack, `v0.23` introduces “splicing”, which allows users to deploy binaries using local, optimized versions of a binary interface, even if they were not built with that interface. For example, this would allow you to build binaries in the cloud using `mpich` and install them on a system using a local, optimized version of `mvapich2` *without rebuilding*. Spack preserves full provenance for the installed packages and knows that they were built one way but deployed another. The intent is to leverage this across many key HPC binary packages, e.g. MPI, CUDA, ROCm, and libfabric. Fundamentally, splicing allows Spack to redeploy an existing spec with different dependencies than how it was built. There are two interfaces to splicing.
 - Explicit Splicing In the concretizer config, you can specify a target spec and a replacement by hash.

```
concretizer:
```

```
splice:
  explicit:
    - target: mpi
      replacement: mpich/abcdef
```

Here, every installation that would normally use the target spec will instead use its replacement. Above, any spec using *any* `mpi` will be spliced to depend on the specific `mpich` installation requested. This *can* go wrong if you try to replace something built with, e.g., `openmpi` with `mpich`, and it is on the user to ensure ABI compatibility between target and replacement specs. This currently requires some expertise to use, but it will allow users to reuse the binaries they create across more machines and environments. . Automatic Splicing (experimental) In the concretizer config, enable automatic splicing:

```
concretizer:
  splice:
    automatic: true
```

or run:

```
spack config add concretizer:splice:automatic:true
```

The concretizer will select splices for ABI compatibility to maximize package reuse. Packages can denote ABI compatibility using the `can_splice` directive. No packages in Spack yet use this directive, so if you want to use this feature you will need to add `can_splice` annotations to your packages. We are working on ways to add more ABI compatibility information to the Spack package repository, and this directive may change in the future. Further documentation: https://spack.readthedocs.io/en/latest/build_settings.html#splicing https://spack.readthedocs.io/en/latest/packaging_guide.html#specifying-abi-compatibility

- **Broader variant propagation** You can specify propagated variants like `hdf5 build_type==RelWithDebInfo` or `trilinos ++openmp` to propagate a variant to all dependencies for which it is relevant. This is valid *even* if the variant does not exist on the package or its dependencies. See https://spack.readthedocs.io/en/latest/basic_usage.html#variants.
- **Query specs by namespace** Allow a package's namespace (indicating the repository it came from) to be treated like a variant. You can request packages from particular repos like this:

```
spack find zlib namespace=builtin
spack find zlib namespace=myrepo
```

Previously, the `spec` syntax only allowed namespaces to be prefixes of spec names, e.g. `builtin.zlib`. The previous syntax still works.

- `spack spec` respects environment settings and `unify:true` `spack spec` did not previously respect environment lockfiles or unification settings, which made it difficult to see exactly how a spec would concretize within an environment. Now it does, so the output you get with `spack spec` will be *the same* as what your environment will concretize to when you run `spack concretize`. Similarly, if you provide multiple specs on the command line with `spack spec`, it will concretize them together if `unify:true` is set.
- Less noisy `spack spec` output `spack spec` previously showed output like this:

```
> spack spec /v5fn6xo
Input spec
-----
-   /v5fn6xo

Concretized
-----
[+]  openssl@3.3.1%apple-clang@16.0.0~docs+shared arch=darwin-sequoia-m1
...
```

But the input spec is redundant, and we know we run `spack spec` to concretize the input spec. `spack spec` now *only* shows the concretized spec.

- Better output for `spack find -c` In an environment, `spack find -c` lets you search the concretized, but not yet installed, specs, just as you would the installed ones. As with `spack spec`, this should make it easier for you to see what *will* be built before building and installing it.
- `spack -C <env>`: use an environment's configuration without activation Spack environments allow you to associate:

1. a set of (possibly concretized) specs, and
2. configuration When you activate an environment, you're using both of these. Previously, we supported:

- `spack -e <env>` to run spack in the context of a specific environment, and
- `spack -C <directory>` to run spack using a directory with configuration files.

You can now also pass an environment to `spack -C` to use *only* the environment's configuration, but not the specs or lockfile.

6.7.3.2 New commands, options, and directives

- The new `spack env track` command takes a non-managed Spack environment and adds a symlink to Spack's `$environments_root` directory, so that it will be included for reference counting for commands like `spack uninstall` and `spack gc`. If you use free-standing directory environments, this is useful for preventing Spack from removing things required by your environments. You can undo this tracking with the `spack env untrack` command.
- Add `-t` short option for `spack --backtrace` `spack -d / --debug` enables backtraces on error, but it can be very verbose, and sometimes you just want the backtrace. `spack -t / --backtrace` provides that option.
- `gc`: restrict to specific specs If you only want to garbage-collect specific packages, you can now provide them on the command line. This gives users finer-grained control over what is uninstalled.
- oci buildcaches now support `--only=package`. You can now push *just* a package and not its dependencies to an OCI registry. This allows dependents of non-redistributable specs to be stored in OCI registries without an error.

6.7.3.3 Highlighted bugfixes

- Externals no longer override the preferred provider. External definitions could interfere with package preferences. Now, if `openmpi` is the preferred `mpi`, and an external `mpich` is defined, a new `openmpi` *will* be built if building it is possible. Previously we would prefer `mpich` despite the preference.
- Composable `cflags`. This release fixes a longstanding bug that concretization would fail if there were different `cflags` specified in `packages.yaml`, `compilers.yaml`, or on the CLI. Flags and their ordering are now tracked in the concretizer and flags from multiple sources will be merged.
- Fix concretizer Unification for included environments.

6.7.3.4 Deprecations, removals, and syntax changes

- The old concretizer has been removed from Spack, along with the `config:concretizer` config option. Spack will emit a warning if the option is present in user configuration, since it now has no effect. Spack now uses a simpler bootstrapping mechanism, where a JSON prototype is tweaked slightly to get an initial concrete spec to download.
- Best-effort expansion of spec matrices has been removed. This feature did not work with the “new” ASP-based concretizer, and did not work with `unify: True` or `unify: when_possible`. Use the `exclude` key (<https://spack.readthedocs.io/en/latest/environments.html#spec-matrices>)[↗] for the environment to exclude invalid components, or use multiple spec matrices to combine the list of specs for which the constraint is valid and the list of specs for which it is not.
- The old Cray `platform` (based on Cray PE modules) has been removed, and `platform=cray` is no longer supported. Since `v0.19`, Spack has handled Cray machines like Linux clusters with extra packages, and we have encouraged using this option to support Cray. The new approach allows us to correctly handle Cray machines with non-SLES operating systems, and it is much more reliable than making assumptions about Cray modules. See the `v0.19` release notes for more details.
- The `config:install_missing_compilers` config option has been deprecated, and it is a no-op when set in `v0.23`. Our new compiler dependency model will replace it with a much more reliable and robust mechanism in `v1.0`.
- Config options that deprecated in `v0.21` have been removed in `v0.23`. You can now only specify preferences for `compilers`, `targets`, and `providers` globally via the `packages:all:` section. Similarly, you can only specify `versions:` locally for a specific package.
- Spack’s old test interface has been removed, having been deprecated in `v0.22.0`. All `builtin` packages have been updated to use the new interface. See the [stand-alone test documentation](https://spack.readthedocs.io/en/latest/packaging_guide.html#stand-alone-tests) (https://spack.readthedocs.io/en/latest/packaging_guide.html#stand-alone-tests)[↗]
- The `spack versions --safe-only` option, deprecated since `v0.21.0`, has been removed.
- The `--dependencies` and `--optimize` arguments to `spack ci` have been deprecated.

6.7.3.5 Binary caches

- Public binary caches now include an ML stack for Linux/aarch64. We now build an ML stack for Linux/aarch64 for all pull requests and on develop. The ML stack includes both CPU-only and CUDA builds for Horovod, Hugging Face, JAX, Keras, PyTorch, scikit-learn, TensorBoard, and TensorFlow, and related packages. The CPU-only stack also includes XGBoost. See <https://cache.spack.io/tag/develop/?stack=ml-linux-aarch64-cuda> .

6.7.3.6 Architecture support

- archspec has been updated to v0.2.5, with support for zen5.
- Spack's CUDA package now supports the Grace Hopper 9.0a compute capability.

6.7.3.7 Other notable changes

- Bugfix: `spack find -x` in environments.
- Spec splices are now robust to duplicate nodes with the same name in a spec.
- Cache per-compiler libc calculations for performance.
- Fixed a bug in external detection for openmpi.
- Mirror configuration allows username/password as environment variables.
- Default library search caps maximum depth/
- Unify interface for `spack spec` and `spack solve` commands.
- Spack no longer RPATHs directories in the default library search path.
- Improved performance of Spack database.
- Enable package reuse for packages with versions from git refs.
- Improved tracking of task queueing/requeueing in the installer.

6.7.4 v0.22.2

6.7.4.1 Bugfixes

- Forward compatibility with Spack 0.23 packages with language dependencies.
- Forward compatibility with `urllib` from Python 3.12.6+.
- Bump vendored `archspec` for better aarch64 support.
- Fix regression in `\\{variants.X}` and `\\{variants.X.value}` format strings.
- Ensure shell escaping of environment variable values in load and activate commands.
- Fix an issue where `spec[pkg]` considers specs outside the current DAG.
- Do not halt concretization on unknown variants in externals.
- Improve validation of `develop` config section/
- Explicitly disable `ccache` if turned off in config, to avoid cache pollution.
- Improve backwards compatibility in `include_concrete`.
- Fix issue where package tags were sometimes repeated.
- Make `setup-env.sh` “sourced only” by dropping execution bits.
- Make certain source/binary fetch errors recoverable instead of a hard error.
- Remove debug statements in package hash computation.
- Remove redundant clingo warnings.
- Remove hard-coded layout version.
- Do not initialize previous store state in `use_store`.

6.7.4.2 Package updates

- chapel major update/

6.7.5 v0.22.1

6.7.5.1 Bug Fixes

- Fix reuse of externals on Linux.
- Ensure parent gcc-runtime version $\geq$ child.
- Ensure the latest gcc-runtime is rpath'ed when multiple exist among link deps.
- Improve version detection of glibc.
- Improve heuristics for solver.
- Make strong preferences override reuse.
- Reduce verbosity when C compiler is missing.
- Make missing ccache executable an error when required.
- Make every environment view containing python a venv.
- Fix external detection for compilers with os but no target.
- Fix version optimization for roots.
- Handle common implementations of pagination of tags in OCI build caches.
- Apply fetched patches to develop specs.
- Avoid Windows wrappers for filesystem utilities on non-Windows.
- Fix formatting issue in spack audit.

6.7.5.2 Package updates

- Require libiconv for iconv. Notice that glibc/musl also provide iconv, but are not guaranteed to be complete. Set packages:iconv:require:[glibc] to restore the old behavior.
- protobuf: fix 3.4:3.21 patch checksum.

- protobuf: update hash for patch needed when = “@3.4:3.21”.
- git: bump v2.39 to 2.45; deprecate unsafe versions.
- gcc: use `-rpath \{rpath_dir\}` not `-rpath=\{rpath_dir\}`.
- Remove mesa18 and libosmesa.
- Enforce consistency of `gl` providers.
- py-matplotlib: qualify when to do a post install.
- rust: fix v1.78.0 instructions.
- suite-sparse: improve setting of the `libs` property.
- netlib-lapack: provide blas and lapack together.

6.8 mpich

`mpich` has been updated to version 3.4.2. For more information see <https://www.mpich.org/2021/05/28/mpich-3-4-2-released/>.

6.9 Slurm

6.9.1 Deprecation of old Versions of Slurm

SLE receives a new Slurm version for roughly every second upstream release. To facilitate this, old version of Slurm will go out of maintenance successively. Users of these versions are encouraged to migrate to a later version before the expiry date. Note, that Slurm only allows migration two versions upwards without loss of data. To migrate to the latest version, migrations to intermediate versions may be required.

Once Slurm versions are out of maintenance, updates to packages depending on this Slurm version will no longer be provided. In particular, this will be the case for `pdsh-slurm` - the Slurm plugin to Pdsh. Note that `pdsh-slurm` is compatible to the Slurm initially shipped with a product/service pack) while packages `pdsh-slurm_<slurm_version>` is compatible with an upgrade version of Slurm.

TABLE 1: TABLE SUNSET SCHEDULE FOR SLURM

Slurm Version	Released for Service Pack	Support End Date
---------------	---------------------------	------------------

17.02	SLE-12-SP2	May 2024
18.08	SLE-15-SP1 and older	October 2024
20.02	SLE-15-SP2 and older	Januar 2025
20.11	SLE-15-SP3/4 and older	January 2026
22.05	SLE-15-SP4 and older	December 2026
23.02	SLE-15-SP5/6 and older	January 2028

6.9.2 Slurm 24.11

6.9.2.1 Important Notes on Upgrading Slurm from a Previous Version

Slurm can be upgraded from version 23.02, 23.11 or 24.05 to version 24.11 without loss of jobs or other state information. Upgrading directly from an earlier version of Slurm will result in loss of state information.

IMPORTANT NOTES: If using the `slurmdbd` (Slurm DataBase Daemon) you must update this first.

The 24.11 `slurmdbd` will work with Slurm daemons of version 23.02 and above. You will not need to update all clusters at the same time, but it is very important to update `slurmdbd` first and having it running before updating any other clusters making use of it.



Note

If using a backup DBD you must start the primary first to do any database conversion, the backup will not start until this has happened.

All SPANK plugins must be recompiled when upgrading from any Slurm version prior to 24.11.

6.9.2.2 Highlights

- Add report `AccountUtilizationByQOS` to `sreport`.
- `AccountUtilizationByUser` is able to be filtered by QOS.

- Add autodetected gpus to the output of `slurmd -C`
- Add ability to submit jobs with multiple QOS. These are sorted by priority highest being the first.
- Removed the instant on feature from `switch/hpe_slingshot`.
- `slurmctld` - Changed incoming RPC handling to dedicated thread pool with asynchronous handling of I/O that can be configured via `conmgr_*` entries under `SlurmctldParameters` in `slurm.conf`.

6.9.2.3 Configuration File Changes (see appropriate man page for details)

- Add `SchedulerParameters=bf_allow_magnetic_slot` option. It allows jobs in magnetic reservations to be planned by backfill scheduler.
- Add `TopologyParam=TopoMaxSizeUnroll=#` to allow `--nodes=<min>-<max>` for `topology/block`.
- Add `DataParserParameters` `slurm.conf` parameter to allow setting default value for CLI `--json` and `--yaml` arguments.
- Hardware collectives in `switch/hpe_slingshot` now requires `enable_stepmgr`.
- Added connection related parameters to `slurm.conf` under `SlurmctldParameters`:
 - `conmgr_max_connections`: Defaults to 150 connections.
 - `conmgr_threads`: Defaults to 64 threads for `slurmctld`.
 - `conmgr_use_poll`: Defaults is to use `epoll` in Linux.
 - `conmgr_connect_timeout`: Defaults to `MessageTimeout`.
 - `conmgr_read_timeout`: Defaults to `MessageTimeout`.
 - `conmgr_wait_write_delay`: Defaults to `MessageTimeout`.
 - `conmgr_write_timeout`: Defaults to `MessageTimeout`.
- Add `SlurmctldParameters=ignore_constraint_validation` to ignore `constraint/feature` validation at submission.
- Add `SchedulerParameters=bf_topopt_enable` option to enable experimental hook to control backfill.

6.9.2.4 Command Changes (see man pages for details)

- Remove `srun --cpu-bind=rank`.
- Add `"%b"` as a file name pattern for the array task id modulo 10.
- `sacct` - Respect `--noheader` for `--batch-script` and `--env-vars`.
- Add `sacctmgr ping` command to query status of `slurmdbd`.
- `sbcast` - Add `--nodelist` option to specify where files are transmitted to
- `sbcast` - Add `--no-allocation` option to transmit files to nodes outside of a job allocation.
- `slurmdbd` - Add `-u` option. This is used to determine if restarting the DBD will result in database conversion.
- Remove `salloc --get-user-env`.
- `scontrol` - Add `--json/--yaml` support to `listpids`.
- `scontrol` - Add `liststeps`.
- `scontrol` - Add `listjobs`.
- `scontrol show topo` - Show aggregated block sizes when using topology/block.

6.9.2.5 API Changes

- Remove `burst_buffer/lua` call `slurm.job_info_to_string()`.
- `job_submit/lua` - Add `assoc_qos` attribute to `job_desc` to display all potential QOS's for a job's association.
- `job_submit/lua` - Add `slurm.get_qos_priority()` function to retrieve the given QOS's priority.

6.9.2.6 SLURMRESTD CHANGES

- Removed fields deprecated in the Slurm-23.11 release from v0.0.42 endpoints.
- Removed v0.0.39 plugins.

- Set `data_parser/v0.0.42+prefer_refs` flag to default.
- Add `data_parser/v0.0.42+minimize_refs` flag to inline single referenced schemas in the OpenAPI schema to get default behavior of `data_parser/v0.0.41`.
- Rename v0.0.42 `JOB_INFO` field `minimum_switches` to `required_switches` to reflect the actual behavior.
- Rename v0.0.42 `ACCOUNT_CONDITION` field `association` to `association` (typo).
- Tag `slurmdb/v0.0.42/jobs pid` field deprecated.

Also Check slurm 24.11 changelog for further details.

6.9.2.7 Update to version 24.11.1

- With client commands `MIN_MEMORY` will show `mem_per_tres` if specified.
- Fix errno message about bad constraint.
- `slurmctld` - Fix crash and possible split brain issue if the backup controller handles an `scontrol` reconfigure while in control before the primary resumes operation.
- Fix `stepmgr` not getting dynamic node addrs from the controller.
- `stepmgr` - avoid `Unexpected missing socket` errors.
- Fix `scontrol show steps` with dynamic `stepmgr`.
- Deny jobs using the `R:` option of `--signal` if `PreemptMode=OFF` globally.
- Force jobs using the `R:` option of `--signal` to be preemptable by requeue or cancel only. If `PreemptMode` on the partition or QOS is off or suspend, the job will default to using `PreemptMode=cancel`.
- If `--mem-per-cpu` exceeds `MaxMemPerCPU`, the number of CPUs per task will always be increased even if `--cpus-per-task` was specified. This is needed to ensure each task gets the expected amount of memory.
- Fix compilation issue on openSUSE Leap 15.
- Fix jobs using more nodes than needed when not using `-N`.
- Fix issue with allocation being allocated less resources than needed when using `--gres-flags=enforce-binding`.

- select/cons_tres - Fix errors with MaxCpusPerSocket partition limit. Used CPUs/cores weren't counted properly, nor limiting free ones to avail, when the socket was partially allocated, or the job request went beyond this limit.
- Fix issue when jobs were preempted for licenses even if there were enough licenses available.
- Fix srun ntasks calculation inside an allocation when nodes are requested using a min-max range.
- Print correct number of digits for TmpDisk in sdiag.
- Fix a regression in 24.11 which caused file transfers to a job with sbcast to not join the job container namespace.
- data_parser/v0.0.40 - Prevent a segfault in the slurmrestd when dumping data with v0.0.40 + complex data parser.
- Remove logic to force lowercase GRES names.
- data_parser/v0.0.42 - Prevent the association id from always being dumped as NULL when parsing in complex mode. Instead it will now dump the id. This affects the following endpoints:
 - GET slurmdb/v0.0.42/association
 - GET slurmdb/v0.0.42/associations
 - GET slurmdb/v0.0.42/config
 - Fixed a job requeuing issue that merged job entries into the same SLUID when all nodes in a job failed simultaneously.
 - When a job completes, try to give idle nodes to reservations with the REPLACE flag before allowing them to be allocated to jobs.
 - Avoid expensive lookup of all associations when dumping or parsing for v0.0.42 endpoints.
 - Avoid expensive lookup of all associations when dumping or parsing for v0.0.41 endpoints.
 - Avoid expensive lookup of all associations when dumping or parsing for v0.0.40 endpoints.

- Fix segfault when testing jobs against nodes with invalid gres.
- Fix performance regression while packing larger RPCs.
- job_container/tmpfs - Fix Xauthority file being created outside the container when EntireStepInNS is enabled.
- job_container/tmpfs - Fix spank_task_post_fork not always running in the container when EntireStepInNS is enabled.
- Fix a job potentially getting stuck in CG on permissions errors while setting up X11 forwarding.
- Fix error on X11 shutdown if Xauthority file was not created.
- slurmctld - Fix memory or fd leak if an RPC is received that is not registered for processing.
- Inject OMPI_MCA_orte_precondition_transports when using PMIx. This fixes mpi apps using Intel OPA, PSM2 and OMPI 5.x when run through srun.
- Don't skip the first partition_job_depth jobs per partition.
- Fix gres allocation issue after controller restart.
- Fix issue where jobs requesting CPUs-per-GPU hang in queue.
- switch/hpe_slingshot - Treat HTTP status forbidden the same as unauthorized, allowing for a graceful retry attempt.
- Package fixes: Slurmdbd no longer starts as root so that the systemd RuntimeDirectory option works properly. This requires further adjustments:
 - Use /run/slurmdbd as directory for PID file. This is created when slurmdbd starts by RuntimeDirectory=slurmdbd in the systemd service file (boo#1236928).
 - Upon installation, create empty /var/log/slurmdbd.log owned by user slurm as slurmdbd is not able to create it (boo#1236929).

6.9.3 Slurm 23.02

6.9.3.1 Important Notes on Upgrading Slurm from a Previous Version

If using the `slurmdbd` (Slurm DataBase Daemon) you must update this first.

If using a backup DBD you must start the primary first to do any database conversion, the backup will not start until this has happened.

The 23.02 `slurmdbd` will work with Slurm daemons of version 21.08 and above. You will not need to update all clusters at the same time, but it is very important to update `slurmdbd` first and having it running before updating any other clusters making use of it.

Slurm can be upgraded from version 22.05 to version 23.02 without loss of jobs or other state information. Upgrading directly from an earlier version of Slurm will result in loss of state information.

All `SPANK` plugins must be recompiled when upgrading from any Slurm version prior to 23.02.



Note

PMIx v1.x is no longer supported.

6.9.3.2 Highlights

- `slurmctld` - Add new RPC rate limiting feature. This is enabled through `SlurmctldParameters=rl_enable`, otherwise disabled by default.
- Make `scontrol` reconfigure and sending a `SIGHUP` to the `slurmctld` behave the same. If you were using `SIGHUP` as a 'lighter' `scontrol` reconfigure to rotate logs please update your scripts to use `SIGUSR2` instead.
- Change cloud nodes to show by default. `PrivateData=cloud` is no longer needed.
- `sreport` - Count planned (FKA reserved) time for jobs running in `IGNORE_JOBS` reservations. Previously was lumped into `IDLE` time.
- `job_container/tmpfs` - Support running with an arbitrary list of private mount points (`/tmp` and `/dev/shm` are the default, but not required).
- `job_container/tmpfs` - Set more environment variables in `InitScript`.

- Make all cgroup directories created by Slurm owned by root. This was the behavior in cgroup/v2 but not in cgroup/v1 where by default the step directories ownership were set to the user and group of the job.
- accounting_storage/mysql - change purge/archive to calculate record ages based on end time, rather than start or submission times.
- job_submit/lua - add support for log_user() from slurm_job_modify().
- Run the following scripts in slurmscriptd instead of slurmctld: ResumeProgram, ResumeFailProgram, SuspendProgram, ResvProlog, ResvEpilog, and RebootProgram (only with SlurmctldParameters=reboot_from_controller).
- Only permit changing log levels with srun --slurmd-debug by root or SlurmUser.
- slurmctld will fatal() when reconfiguring the job_submit plugin fails.
- Add PowerDownOnIdle partition option to power down nodes after nodes become idle.
- Add “[jobid.stepid]” prefix from slurmstepd and “slurmscriptd” prefix from slurmscriptd to Syslog logging. Previously was only happening when logging to a file.
- Add purge and archive functionality for job environment and job batch script records.
- Extend support for Include files to all "configless" client commands.
- Make node weight usable for powered down and rebooting nodes.
- Removed “launch” plugin.
- Add “Extra” field to job to store extra information other than a comment.
- Add usage gathering for AMD (requires ROCM 5.5 +) and NVIDIA gpus.
- Add job’s allocated nodes, features, oversubscribe, partition, and reservation to SLURM_RESUME_FILE output for power saving.
- Automatically create directories for stdout/stderr output files. Paths may use %j and related substitution characters as well.
- Add --tres-per-task to salloc/sbatch/srun.
- Allow nodefeatures plugin features to work with cloud nodes. e.g. - Powered down nodes have no active changeable features.

- Nodes can't be changed to other active features until powered down.
- Active changeable features are reset/cleared on power down.
- Make `slurmstepd` cgroups constrained by total configured memory from `slurm.conf` (`NodeName=<> RealMemory=#`) instead of total physical memory.
- `node_features/helpers` - add support for the OR and parentheses operators in a `--constraint` expression.
- `slurmctld` will `fatal()` when `[Prolog|Epilog]Slurmctld` are defined but are not executable.
- Validate node registered active features are a super set of node's currently active changeable features.
- On clusters without any `PrologFlags` options, batch jobs with failed prologs no longer generate an output file.
- Add `SLURM_JOB_START_TIME` and `SLURM_JOB_END_TIME` environment variables.
- Add `SuspendExcStates` option to `slurm.conf` to avoid suspending/powering down specific node states.
- Add support for DCMI power readings in IPMI plugin.
- `slurmrestd` served `/slurm/v0.0.39` and `/slurmdb/v0.0.39` endpoints had major changes from prior versions. Almost all schemas have been renamed and modified. Sites using OpenAPI Generator clients are highly suggested to upgrade to using atleast version 6.x due to limitations with prior versions.
- Allow for `--odelist` to contain more nodes than required by `--nodes`.
- Rename "nodes" to "nodes_resume" in `SLURM_RESUME_FILE` job output.
- Rename "all_nodes" to "all_nodes_resume" in `SLURM_RESUME_FILE` output.
- Add `jobcomp/kafka` plugin.
- Add new `PreemptParameters=reclaim_licenses` option which will allow higher priority jobs to preempt jobs to free up used licenses. (This is only enabled for with `Preempt-Modes` of `CANCEL` and `REQUEUE`, as Slurm cannot guarantee suspended jobs will release licenses correctly.)

- hpe/slingshot - add support for the instant-on feature.
- Add ability to update SuspendExc* parameters with scontrol.
- Add ability to restore SuspendExc* parameters on restart with slurmctld -R option.
- Add ability to clear a GRES specification by setting it to "0" via “scontrol update job”.
- Add SLURM_JOB_OVERSUBSCRIBE environment variable for Epilog, Prolog, EpilogSlurmctld, PrologSlurmctld, and mail output.
- System node down reasons are appended to existing reasons, separated by ' '.
- New command scrunch has been added. scrunch acts as an Open Container Initiative (OCI) runtime proxy to run containers seamlessly via Slurm.
- Fixed GpuFreqDef option. When set in slurm.conf, it will be used if --gpu-freq was not explicitly set by the job step.

6.9.3.3 Configuration File Changes (see appropriate man page for details)

- job_container.conf - Added “Dirs” option to list desired private mount points.
- node_features plugins - invalid users specified for AllowUserBoot will now result in fatal() rather than just an error.
- Deprecate AllowedKmemSpace, ConstrainKmemSpace, MaxKmemPercent, and MinKmemSpace.
- Allow jobs to queue even if the user is not in AllowGroups when EnforcePartLimits=no is set. This ensures consistency for all the Partition access controls, and matches the documented behavior for EnforcePartLimits.
- Add InfluxDBTimeout parameter to acct_gather.conf.
- job_container/tmpfs - add support for expanding %h and %n in BasePath.
- slurm.conf - Removed SlurmctldPlugstack option.
- Add new SlurmctldParameters=validate_nodeaddr_threads=<number> option to allow concurrent hostname resolution at slurmctld startup.

- Add new `AccountingStoreFlags=job_extra` option to store a job's extra field in the database.
- Add new “defer_batch” option to `SchedulerParameters` to only defer scheduling for batch jobs.
- Add new `DebugFlags` option “JobComp” to replace “Elasticsearch”.
- Add configurable job requeue limit parameter - `MaxBatchRequeue` - in `slurm.conf` to permit changes from the old hard-coded value of 5.
- `helpers.conf` - Allow specification of node specific features.
- `helpers.conf` - Allow many features to one helper script.
- `job_container/tmpfs` - Add “Shared” option to support shared namespaces. This allows autofs to work with the `job_container/tmpfs` plugin when enabled.
- `acct_gather.conf` - Added `EnergyIPMIPowerSensors=Node=DCMI` and `Node=D-CMI_ENHANCED`.
- Add new “getnameinfo_cache_timeout= <number>” option to `CommunicationParameters` to adjust or disable caching the results of `getnameinfo()`.
- Add new `PrologFlags=ForceRequeueOnFail` option to automatically requeue batch jobs on Prolog failures regardless of the job `--requeue` setting.
- Add `HealthCheckNodeState=NONDRAINED_IDLE` option.
- Add “explicit” to `Flags` in `gres.conf`. This makes it so the gres is not automatically added to a job's allocation when `--exclusive` is used. Note that this is a per-node flag.
- Moved the “preempt_” options from `SchedulerParameters` to `PreemptParameters`, and dropped the prefix from the option names. (The old options will still be parsed for backwards compatibility, but are now undocumented.)
- Add `LaunchParameters=ulimit_pam_adapt`, which enables setting `RLIMIT_RSS` in adopted processes.
- Update `SwitchParameters=job_vni` to enable/disable creating job VNIs for all jobs, or when a user requests them.
- Update `SwitchParameters=single_node_vni` to enable/disable creating single node VNIs for all jobs, or when a user requests them.

- Add ability to preserve SuspendExc* parameters on reconfig with ReconfigFlags=Keep-PowerSaveSettings.
- slurmdbd.conf - Add new AllResourcesAbsolute to force all new resources to be created with the Absolute flag.
- topology/tree - Add new TopologyParam=SwitchAsNodeRank option to reorder nodes based on switch layout. This can be useful if the naming convention for the nodes does not naturally map to the network topology.
- Removed the default setting for GpuFreqDef. If unset, no attempt to change the GPU frequency will be made if --gpu-freq is not set for the step.

6.9.3.4 Command Changes (see man pages for details)

- sacctmgr - no longer force updates to the AdminComment, Comment, or SystemComment to lower-case.
- sinfo - Add -F/--future option to sinfo to display future nodes.
- sacct - Rename “Reserved” field to “Planned” to match sreport and the nomenclature of the 'Planned' node.
- scontrol - advanced reservation flag MAINT will no longer replace nodes, similar to STATIC_ALLOC
- sbatch - add parsing for #PBS -d and #PBS -w.
- scontrol show assoc_mgr will show username(uid) instead of uid in QoS section.
- Add strigger --draining and -R/--resume options.
- Change --oversubscribe and --exclusive to be mutually exclusive for job submission. Job submission commands will now fatal if both are set. Previously, these options would override each other, with the last one in the job submission command taking effect.
- scontrol - Requested TRES and allocated TRES will now always be printed when showing jobs, instead of one TRES output that was either the requested or allocated.
- srun --ntasks-per-core now applies to job and step allocations. Now, use of --ntasks-per-core=1 implies --cpu-bind=cores and --ntasks-per-core>1 implies --cpu-bind=threads.

- salloc/sbatch/srun - Check and abort if `ntasks-per-core > threads-per-core`.
- scontrol - Add `ResumeAfter=<secs>` option to “`scontrol update nodename=`”.
- Add a new “`nodes=`” argument to scontrol setdebug to allow the debug level on the slurmd processes to be temporarily altered.
- Add a new “`nodes=`” argument to “`scontrol setdebugflags`” as well.
- Make it so scrontab prints client-side the job_submit() error message (which can be set i.e. by using the log_user() function for the lua plugin).
- scontrol - Reservations will not be allowed to have STATIC_ALLOC or MAINT flags and REPLACE[_DOWN] flags simultaneously.
- scontrol - Reservations will only accept one reoccurring flag when being created or updated.
- scontrol - A reservation cannot be updated to be reoccurring if it is already a floating reservation.
- squeue - removed unused “`%s`” and “`SelectJobInfo`” formats.
- squeue - align print format for exit and derived codes with that of other components (`<exit_status>:<signal_number>`).
- sacct - Add `--array` option to expand job arrays and display array tasks on separate lines.
- Partial support for “`--json`” and “`--yaml`” formatted outputs have been implemented for sacctmgr, sdiag, sinfo, squeue, and scontrol. The resultant data output will be filtered by normal command arguments. Formatting arguments will continue to be ignored.
- salloc/sbatch/srun - extended the `--nodes` syntax to allow for a list of valid node counts to be allocated to the job. This also supports a "step count" value (e.g., `--nodes=20-100:20` is equivalent to `--nodes=20,40,60,80,100`) which can simplify the syntax when the job needs to scale by a certain "chunk" size.
- srun - add user requestible vnis with “`--network=job_vni`” option.
- srun - add user requestible single node VNIs with the “`--network=single_node_vni`” option.

6.9.3.5 API Changes

- job_container plugins - container_p_stepd_create() function signature replaced uint32_t uid with stepd_step_rec_t* step.
- gres plugins - gres_g_get_devices() function signature replaced pid_t pid with stepd_step_rec_t* step.
- cgroup plugins - task_cgroup_devices_constrain() function signature removed pid_t pid.
- task plugins - replace task_p_pre_set_affinity(), task_p_set_affinity(), and task_p_post_set_affinity() with task_p_pre_launch_priv() like it was back in slurm 20.11.
- Allow for concurrent processing of job_submit_g_submit() and job_submit_g_modify() calls. If your plugin is not capable of concurrent operation you must add additional locking within your plugin.
- Removed return value from slurm_list_append().
- The List and ListIterator types have been removed in favor of list_t and list_itr_t respectively.
- burst_buffer plugins - add bb_g_build_het_job_script(). bb_g_get_status() - added authenticated UID and GID. bb_g_run_script() - added job_info argument.
- burst_buffer.lua - Pass UID and GID to most hooks. Pass job_info (detailed job information) to many hooks. See etc/burst_buffer.lua.example for a complete list of changes. WARNING: Backwards compatibility is broken for slurm_bb_get_status: UID and GID are passed before the variadic arguments. If UID and GID are not explicitly listed as arguments to slurm_bb_get_status(), then they will be included in the variadic arguments. Backwards compatibility is maintained for all other hooks because the new arguments are passed after the existing arguments.
- node_features plugins - node_features_p_reboot_weight() function removed. node_features_p_job_valid() - added parameter feature_list. node_features_p_job_xlate() - added parameters feature_list and job_node_bitmap.
- New data_parser interface with v0.0.39 plugin.

6.9.4 Important Notes for Upgrading Slurm Releases:

If using the slurmdbd (Slurm DataBase Daemon) you must update this first. If using a backup DBD you must start the primary first to do any database conversion, the backup will not start until this has happened.

6.9.5 Slurm version 22.05

An update to Slurm version 22.05 is available.

6.9.5.1 Important notes for upgrading to version 22.05

Slurmdbd version 22.05 will work Slurm daemons of version 20.11. You will not need to update all clusters at the same time, but it is very important to update slurmdbd first and having it running before updating any other clusters making use of it.

Slurm can be upgraded from version 20.11 to version 22.05 without loss of jobs or other state information. Upgrading directly from an earlier version of Slurm will result in loss of state information.

For more information and a recommended upgrade procedure, see the section "Upgrading Slurm" in the chapter "Slurm — utility for HPC workload management" of the in the SLE HPC 15 "Administration Guide".

All SPANK plugins must be recompiled when upgrading from any Slurm version prior to 22.05. If you are using the Slurm plugin for pdsh you must make sure, pdsh_slurm_22_05 is installed together with slurm_22_05.

6.9.5.2 Highlights of version 20.11

- The template slurmrestd.service unit file now defaults to listen on both the Unix socket and the slurmrestd port.
- The template slurmrestd.service unit file now defaults to enable auth/jwt and the munge unit is no longer a dependency by default.
- Add extra “EnvironmentFile = -/etc/default/\$service” setting to service files.
- Allow jobs to pack onto nodes already rebooting with the desired features.

- Reset job start time after nodes are rebooted, previously only done for cloud/power save boots.
- Node features (if any) are passed to `RebootProgram` if run from `slurmctld`.
- Fail `srun` when using invalid `--cpu-bind` options (e.g. `--cpu-bind=map_cpu:99` when only 10 CPUs are allocated).
- Storing batch scripts and env vars are now in indexed tables using substantially less disk space. Those storing scripts in 21.08 will all be moved and indexed automatically.
- Run `MailProg` through `slurmscriptd` instead of directly `fork+exec()`ing from `slurmctld`.
- Add `acct_gather_interconnect/sysfs` plugin.
- Future and Cloud nodes are treated as "Planned Down" in usage reports.
- Add new shard plugin for sharing GPUs but not with mps.
- Add support for Lenovo SD650 V2 in `acct_gather_energy/xcc` plugin.
- Remove `cgroup_allowed_devices_file.conf`, since the default policy in modern kernels is to whitelist by default. Denying specific devices must be done through `gres.conf`.
- Node state flags (`DRAIN`, `FAILED`, `POWERING UP`, etc.) will be cleared now if node state is updated to `FUTURE`.
- `srun` will no longer read in `SLURM_CPUS_PER_TASK`. This means you will implicitly have to specify `--cpus-per-task` on your `srun` calls, or set the new `SRUN_CPUS_PER_TASK` environment variable to accomplish the same thing.
- Remove `connect_timeout` and `timeout` options from `JobCompParams` as there's no longer a connectivity check happening in the `jobcomp/elasticsearch` plugin when setting the location off of `JobCompLoc`.
- Add support for hourly reoccurring reservations.
- Allow nodes to be dynamically added and removed from the system. Configure `MaxNodeCount` to accomodate nodes created with dynamic node registrations (`slurmd -Z --conf=""`) and `scontrol`.
- Added support for Cgroup Version 2.

- sacct - allocations made by srun will now always display the allocation and step(s). Previously, the allocation and step were combined when possible.
- cons_tres - change definition of the "least loaded node" (LLN) to the node with the greatest ratio of available CPUs to total CPUs.
- Add support to ship Include configuration files with configless.
- Provide a detailed reason in the job log as to why it has been terminated when hitting a resource limit.
- Pass and use alias_list through credential instead of environment variable.
- Add ability to get host addresses from nss_slurm.
- Enable reverse fanout for cloud+alias_list jobs.
- Add support to delete/update nodes by specifying nodesets or the 'ALL' keyword alongside the delete/update node message nodelist expression (i.e. scontrol delete/update NodeName=ALL or scontrol delete/update NodeName=ns1,nodes[1-3]).
- Expanded the set of environment variables accessible through Prolog/Epilog and PrologSlurmctld/EpilogSlurmctld to include SLURM_JOB_COMMENT, SLURM_JOB_STDERR, SLURM_JOB_STDIN, SLURM_JOB_STDOUT, SLURM_JOB_PARTITION, SLURM_JOB_ACCOUNT, SLURM_JOB_RESERVATION, SLURM_JOB_CONSTRAINTS, SLURM_JOB_NUM_HOSTS, SLURM_JOB_CPUS_PER_NODE, SLURM_JOB_NTASKS, and SLURM_JOB_RESTART_COUNT.
- Attempt to requeue jobs terminated by slurm.conf changes (node vanish, node socket/core change, etc). Processes may still be running on excised nodes. Admin should take precautions when removing nodes that have jobs on running on them.
- Add switch/hpe_slingshot plugin.
- Add new SchedulerParameters option bf_licenses to track licenses as within the backfill scheduler.

6.9.5.3 Configuration File changes (for details, see the appropriate man page)

- AcctGatherEnergyType rsmi is now gpu.
- TaskAffinity parameter was removed from cgroup.conf.

- Fatal if the mutually-exclusive `JobAcctGatherParams` options of `UsePss` and `NoShared` are both defined.
- `KeepAliveTime` has been moved into `CommunicationParameters`. The standalone option will be removed in a future version.
- `preempt/qos` - add support for WITHIN mode to allow for preemption between jobs within the same QOS.
- Fatal error if `CgroupReleaseAgentDir` is configured in `cgroup.conf`. The option has long been obsolete.
- Fatal if more than one burst buffer plugin is configured.
- Added `keepaliveinterval` and `keepaliveprobes` to `CommunicationParameters`.
- Added new `max_token_lifespan=<seconds>` to `AuthAltParameters` to allow sites to restrict the lifespan of any requested ticket by an unprivileged user.
- Disallow `slurm.conf` node configurations with `NodeName=ALL`.

6.9.5.4 Command Changes (for details, see the appropriate man page)

- Remove support for (non-functional) `--cpu-bind=boards`.
- Added `--prefer` option at job submission to allow for 'soft' constraints.
- Add `condflags=open` to `sacctmgr show events` to return open/currently down events.
- `sacct -f` flag implies `-c` flag.
- `srun --overlap` now allows the step to share all resources (CPUs, memory, and GRES), where previously `--overlap` only allowed the step to share CPUs with other steps.

6.9.5.5 API Changes

- `openapi/v0.0.35` - Plugin has been removed.
- `burst_buffer` plugins - `err_msg` added to `bb_p_job_validate()`.
- `openapi` - added flags to `slurm_openapi_p_get_specification()`. Existing plugins only need to update their prototype for the function as manipulating the flags pointer is optional.

- openapi - Added OAS_FLAG_MANGLE_OPID to allow plugins to request that the operationId of path methods be mangled with the full path to ensure uniqueness.
- openapi/[db]v0.0.36 - Plugins have been marked as deprecated and will be removed in the next major release.
- switch plugins - add switch_g_job_complete() function.

6.9.6 Highlights of Slurm version 21.08

6.9.6.1 Highlights

- Removed gres/mic plugin used to support Xeon Phi coprocessors.
- Add LimitFactor to the QOS. A float that is factored into an associations GrpTRES limits. For example, if the LimitFactor is 2, then an association with a GrpTRES of 30 CPUs, would be allowed to allocate 60 CPUs when running under this QOS.
- A job's next_step_id counter now resets to 0 after being requeued. Previously, the step id's would continue from the job's last run.
- API change: Removed slurm_kill_job_msg and modified the function signature for slurm_kill_job2. slurm_kill_job2 should be used instead of slurm_kill_job_msg.
- AccountingStoreFlags=job_script allows you to store the job's batch script.
- AccountingStoreFlags=job_env allows you to store the job's env vars.
- Removed sched/hold plugin.
- cli_filter/lua, jobcomp/lua, job_submit/lua now load their scripts from the same directory as the slurm.conf file (and thus now will respect changes to the SLURM_CONF environment variable).
- SPANK - call slurm_spank_init if defined without slurm_spank_slurmd_exit in slurmd context.
- Add new PLANNED state to a node to represent when the backfill scheduler has it planned to be used in the future instead of showing as IDLE. sreport also has changed it's cluster utilization report column name from 'Reserved' to 'Planned' to match this nomenclature.

- Put node into INVALID state upon registering with an invalid node configuration. Node must register with a valid configuration to continue.
- Remove SLURM_DIST_LLLP environment variable in favor of just SLURM_DISTRIBUTION.
- Make --cpu-bind=threads default for --threads-per-core — can be overridden by the CLI or an environment variable.
- slurmd - allow multiple comma-separated controllers to be specified in configless mode with --conf-server
- Manually powering down of nodes with scontrol now ignores SuspendExc<Nodes|Parts>.
- Distinguish queued reboot requests (REBOOT@) from issued reboots (REBOOT^).
- auth/jwt - add support for RS256 tokens. Also permit the username in the 'username' field in addition to the 'sun' (Slurm UserName) field.
- service files - change dependency to network-online rather than just network to ensure DNS and other services are available.
- Add "Extra" field to node to store extra information other than a comment.
- Add ResumeTimeout, SuspendTimeout and SuspendTime to Partitions.
- The memory.force_empty parameters is no longer set by jobacct_gather/cgroup when deleting the cgroup`. This previously caused a significant delay (~2s) when terminating a job, and is not believed to have provided any perceivable benefit. However, this may lead to slightly higher reported kernel mem page cache usage since the kernel cgroup memory is no longer freed immediately.
- TaskPluginParam=verbose is now treated as a default. Previously it would be applied regardless of the job specifying a --cpu-bind.
- Add node_reg_mem_percent SlurmctldParameter to define percentage of memory nodes are allowed to register with.
- Define and separate node power state transitions. Previously a powering down node was in both states, POWERING_OFF and POWERED_OFF. These are now separated. e.g. IDLE+POWERED_OFF (IDLE~) → IDLE+POWERING_UP (IDLE#) - Manual power up or allocation → IDLE → IDLE+POWER_DOWN (IDLE!) - Node waiting for power down → IDLE+POWERING_DOWN (IDLE%) - Node powering down → IDLE+POWERED_OFF (IDLE~) - Powered off

- Some node state flag names have changed. These would be noticeable for example if using a state flag to filter nodes with `sinfo`. e.g. `POWER_UP` → `POWERING_UP` `POWER_DOWN` → `POWERED_DOWN` `POWER_DOWN` now represents a node pending power down
- Create a new process called `slurmscriptd` which runs `PrologSlurmctld` and `Epi-logSlurmctld`. This avoids `fork()` calls from `slurmctld`, and can avoid performance issues if the `slurmctld` has a large memory footprint.
- Pass JSON of job to node mappings to `ResumeProgram`.
- QOS accrue limits only apply to the job QOS, not partition QOS.
- Any return code from SPANK plugin or SPANK function that is not `SLURM_SUCCESS` (zero) will be considered to be an error. Previously, only negative return codes were considered an error.
- Add support for automatically detecting and broadcasting executable shared object dependencies for `sbcst` and `srun --bcst`.
- All SPANK error codes now start at 3000. Where previously SPANK would give a return code of 1, it will now return 3000. This change will break ABI compatibility with SPANK plugins compiled against older version of Slurm.
- SPANK plugins are now required to match the current Slurm release, and must be recompiled for each new Slurm major release. (They do not need to be recompiled when upgrading between maintenance releases.)
- `SLURM_NODE_ALIASES` now has brackets around the node's address to be able to distinguish IPv6 addresses. e.g. `<node_name>:[<node_addr>]:<node_hostname>`
- The `job_container/tmpfs` plugin now requires `PrologFlags=contain` to be set in `slurm.conf`.
- Limit `max_script_size` to 512 MB.

6.9.6.2 Configuration File Changes (for details, see the appropriate man page)

- Errors detected in the parser handlers due to invalid configurations are now propagated and can lead to fatal (and thus exit) the calling process.
- Enforce a valid configuration for `AccountingStorageEnforce` in `slurm.conf`. If the configuration is invalid, then an error message will be printed and the command or daemon (including `slurmctld`) will not run.

- Removed AccountingStoreJobComment option. Please update your config to use AccountingStoreFlags=job_comment instead.
- Removed DefaultStorage{Host,Loc,Pass,Port,Type,User} options.
- Removed CacheGroups, CheckpointType, JobCheckpointDir, MemLimitEnforce, SchedulerPort, SchedulerRootFilter options.
- Added Script to DebugFlags for debugging slurmscriptd (the process that runs slurmctld scripts such as PrologSlurmctld and EpilogSlurmctld).
- Rename SbcastParameters to BcastParameters.
- systemd service files - add new “-s” option to each daemon which will change the working directory even with the -D option. (Ensures any core files are placed in an accessible location, rather than /.)
- Added BcastParameters=send_libs and BcastExclude options.
- Remove the (incomplete) burst_buffer/generic plugin.
- Make SelectTypeParameters=CR_Core_Memory default for cons_tres and cons_res.
- Remove support for TaskAffinity=yes in cgroup.conf. Adding task/affinity to TaskPlugins in slurm.conf is strongly recommended instead.

6.9.6.3 Command Changes (for details, see the appropriate man page)

- Changed the --format handling for negative field widths (left justified) to apply to the column headers as well as the printed fields.
- Invalidate multiple partition requests when using partition based associations.
- scrontab - create the temporary file under the TMPDIR environment variable (if set), otherwise continue to use TmpFS as configured in slurm.conf.
- sbcast / srun --bcast - removed support for zlib compression. lz4 is vastly superior in performance, and (counter-intuitively) zlib could provide worse performance than no compression at all on many systems.
- sacctmgr - changed column headings to ParentID and ParentName instead of Par ID and "Par Name` respectively.

- `SALLOC_THREADS_PER_CORE` and `SBATCH_THREADS_PER_CORE` have been added as input environment variables for `salloc` and `sbatch`, respectively. They do the same thing as `--threads-per-core`.
- Don't display node's comment with `scontrol show nodes` unless set.
- Added `SLURM_GPUS_ON_NODE` environment variable within each job/step.
- `sreport` - change to sorting `TopUsage` by the `--tres` option.
- `slurmrestd` - do not run allow operation as `SlurmUser`/root by default.
- `scontrol show node` now shows `State` as `base_state+flags` instead of shortened state with flags appended. eg. `IDLE#` → `IDLE+POWERING_UP`. Also `POWER` state flag string is `POWERED_DOWN`.
- `scrontab` - add ability to update crontab from a file or standard input.
- `scrontab` - added ability to set and expand variables.
- Make `srun` sensitive to `BcastParameters`.
- Added `sbcast/srun --send-libs`, `sbcast --exclude` and `srun --bcast-exclude`.
- Changed `ReqMem` field in `sacct` to match memory from `ReqTRES`. It now shows the requested memory of the whole job with a letter appended indicating units (M for megabytes, G for gigabytes, etc.). `ReqMem` is only displayed for the job, since the step does not have requested TRES. Previously `ReqMem` was also displayed for the step but was just displaying `ReqMem` for the job.

6.9.6.4 API Changes

- `jobcomp` plugin: change plugin API to `jobcomp_p_*`.
- `sched` plugin: change plugin API to `sched_p_*` and remove `slurm_sched_p_initial_priority()` call.
- `step_ctx` code has been removed from the api.
- `slurm_stepd_get_info()` / `stepd_get_info()` has been removed from the api.
- The v0.0.35 OpenAPI plugin has now been marked as deprecated. Please convert your requests to the v0.0.37 OpenAPI plugin.

6.10 Creating containers from current HPC environment

Usually users use environment modules to adjust their environment (that is, environment variables like `PATH`, `LD_LIBRARY_PATH`, `MANPATH` etc.) to pick exactly the tools and libraries they need for their work. The same can be achieved with containers by including only those components in a container that are part of this environment. This functionality is now provided using the `spack` and `singularity` applications.

7 Removed and deprecated features and packages

This section lists features and packages that were removed from SUSE Linux Enterprise for High-Performance Computing or will be removed in upcoming versions.

7.1 Removed features and packages

The following features and packages have been removed in this release.

- Python 2 bindings for `genders` has been removed. These are now provided for Python 3.
- Ganglia is not supported anymore in 15 SP4. It has been replaced with Grafana (<https://grafana.com/> )
- Due to a lack of usage by customers, some library packages have been removed from the HPC module in SLE HPC 15 SP4. On SUSE Linux Enterprise you can build your own library using `spack`. These libraries will continue to be available through SUSE Package Hub. The following libraries have been removed:

- `boost`
- `adios`
- `gsl`
- `fftw3`
- `hypre`
- `metis`
- `mumps`

- [netcdf](#)
- [ocr](#)
- [petsc](#)
- [ptscotch](#)
- [scalapack](#)
- [superlu](#)
- [trilinos](#)

7.2 Deprecated features and packages

The following features and packages are deprecated and will be removed in a future version of SUSE Linux Enterprise for High-Performance Computing.

8 Obtaining source code

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A Changelog for 15 SP4

A.1 2025-10-31

A.1.1 New

- *Section 6.1, "SLE HPC no longer a separate product"* (Jira (<https://jira.suse.com/browse/PED-7684>) )
- *Section 6.9.3, "Slurm 23.02"* (Jira (<https://jira.suse.com/browse/PED-2802>) )
- *Section 6.7, "spack"* (Jira (<https://jira.suse.com/browse/PED-2803>) )

A.2 2022-11-30

A.2.1 New

- Added *Section 6.2, "Enriched system visibility in the SUSE Customer Center (SCC)"* (Jira (<https://jira.suse.com/browse/SLE-24988>) )

A.3 2022-08-31

A.3.1 New

- Added *Section 6.3, "Automatically opened ports"* (Jira (<https://jira.suse.com/browse/SLE-22743>) )

A.4 2022-05-11

A.4.1 New

- Added this changelog

A.5 2022-03-23

A.5.1 New

- Added *Section 6.9, "Slurm"* (Jira (<https://jira.suse.com/browse/SLE-12352>) )
- Added notes about dolly, memkind, openblas, spack, and mpich in *Section 6, "Changes affecting all architectures"*
- Added note about Ganglia being unsupported in *Section 7, "Removed and deprecated features and packages"* (Jira (<https://jira.suse.com/browse/SLE-17777>) )
- Added note about removal of Python 2 bindings for genders (Jira (<https://jira.suse.com/browse/SLE-23359>) )

A.5.2 Updates

- Added a note about building libraries using spack in *Section 7, “Removed and deprecated features and packages”* (Jira (<https://jira.suse.com/browse/SLE-17776>) )
- Added adios and superlu to the list of removed libraries in *Section 7, “Removed and deprecated features and packages”*

A.6 2021-11-03

- Initial SP4 release